

Project Title: Kuku AI Companion — Smart Visual & Voice Enhancer for Audiobooks

Objective:

To minimize distraction and enhance user comprehension by integrating generative AI visuals, smart caching, personalized summaries, and celebrity voice narration into the Kuku FM audiobook experience. This feature helps users stay immersed in the story and improves memory retention by providing relevant AI-generated visuals that mirror the audio content. The result is a more engaging, memorable, and personalized audiobook journey.

User Persona:

Ravi Verma, a 22-year-old Computer Science student from Pune, is passionate about self-improvement and enjoys listening to audiobooks on Kuku FM during his daily metro commute and evening walks. However, he often finds it difficult to focus for extended periods due to external distractions and multitasking. Ravi is a visual learner—he remembers better when he can associate concepts with images or scenes. He also prefers listening in familiar voices that he emotionally connects with. He wishes there were a smarter way to stay engaged with the books and pick up right where he left off, without re-listening or losing track of the story.

Key Features:

1. AI-Generated Visuals & Animations

- AI creates real-time or pre-cached images or short animations that reflect the audiobook content.
- Enhances engagement and comprehension for visual learners.
- Especially useful for self-help, biographies, fiction, and educational genres.

2. Smart Cache + Resume Feature

- Automatically saves user progress and listening checkpoints.
- Resumes content from the exact stopping point.
- Offers quick summaries of previously listened content using AI-generated visuals and audio/text.

3. Celebrity Voice Customization

- Users can choose from a library of licensed celebrity voices for narration.
- Creates a highly personalized and premium listening experience.
- A strong incentive for Kuku Premium subscriptions.



Technology Stack & AI Integration:

Feature	Tools/Models
Visual Generation	Stable Diffusion, DALL·E, Runway ML
Content Summarization	GPT-4 fine-tuned on Kuku’s content
Resume & Smart Caching	Timestamp + Chapter Bookmarks
Voice Customization	ElevenLabs, Respeecher (Voice Cloning)

Workflow:

1. User starts audiobook.
2. Visuals auto-generate based on ongoing audio (preloaded or live).
3. User pauses/stops → app saves position + caches visual/audio summary.
4. On returning: quick summary presented with resume option.
5. Optional: switch to preferred celebrity voice.

Implementation Roadmap:

Phase	Deliverables
Phase 1	Visual Companion + Smart Resume + Summary
Phase 2	Celebrity Voice Integration (Premium Tier)
Phase 3	Regional Voice Packs + Multilingual Voices

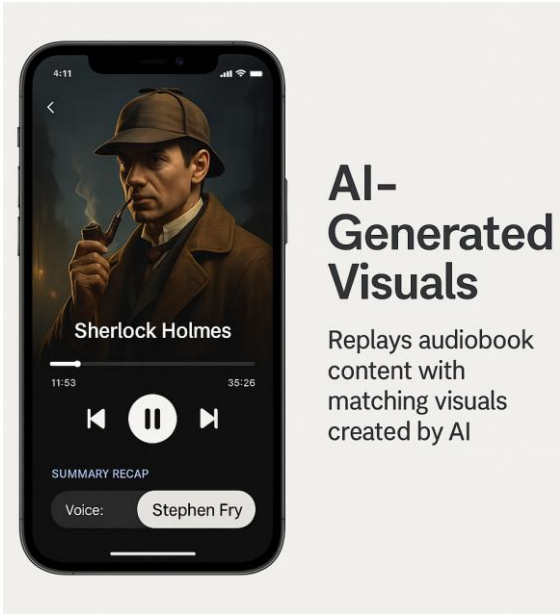
Competitive Advantage:

While Spotify and Audible offer basic audio playback and bookmarks, Kuku AI Companion introduces:

- Live or cached **AI-generated visuals** for better comprehension.
- A **personalized visual/audio summary** experience.
- **Celebrity voice cloning**, providing emotional relatability and uniqueness.

This makes Kuku FM a more immersive and user-focused platform.

Success Metrics (KPIs):



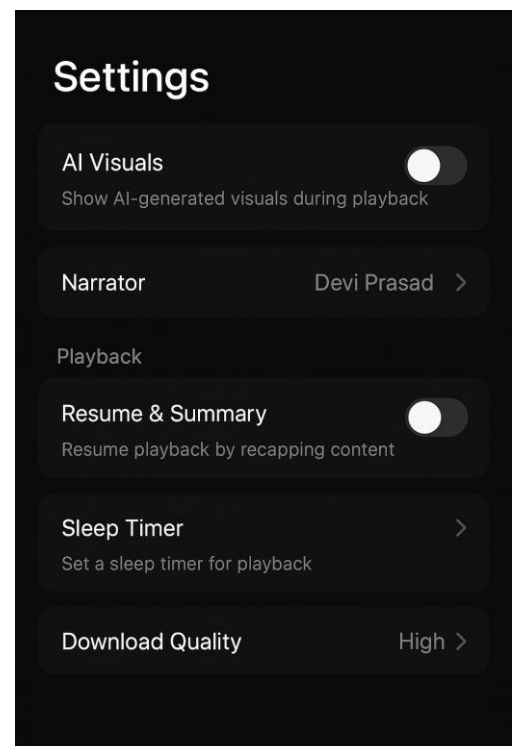
Metric	Target
Avg. Session Duration	+40%
Repeat App Opens per Day	+30%
Audiobook Completion Rate	+25%
Summary Feature Usage	>60% of users
Celebrity Voice Subscription	10–15% growth

Challenges & Solutions:

Challenge	Proposed Solution
Real-time visual generation	Pre-render visuals using keyword chunking
Celebrity voice licensing	Partner with licensed providers or creators
App performance on low-end devices	Offer "Light Mode" with minimal visuals

Mockup Visual Concepts:

- Visual Playback Screen:**
 - Audiobook with animated background visuals
 - Option to toggle between voice narrators
- Resume + Summary Screen:**
 - Timeline of previous listening sessions
 - Quick recap (text + audio + visuals)
- Voice Selection Screen:**
 - Grid of celebrity avatars or voices to select from
 - Preview option + Premium subscription CTA
- Settings Page:**
 - Enable/disable visuals
 - Voice options
 - Summary playback toggle



The Kuku AI Companion will transform the traditional audiobook experience into an immersive, emotionally engaging, and personalized journey. By addressing user drop-offs and increasing cognitive retention, this feature has the potential to significantly boost user engagement, satisfaction, and subscription conversion rates.

Which Type of Model to Use?

Deep Learning (Recommended First Step)

- Use **Audio → Text → Image** pipeline:
 1. **ASR (Automatic Speech Recognition):**
 - Convert audiobook audio to text using pretrained models (e.g., [Whisper](#) by OpenAI).
 2. **Text-to-Image Generator:**
 - Use models like **Stable Diffusion** or **DALL·E** to generate visuals from short summaries or captions.

Reinforcement Learning (Advanced Idea)

- Use RL if you want the model to learn which visuals increase **engagement** or **recall** based on feedback.
- Can build an RL agent that:
 - Gets **rewarded** for generating visuals users engage with
 - Learns from user behavior over time

 This is cutting-edge and complex. Not necessary for MVP but great for future scope.

Prototype Plan

1. **Input:** Sample audiobook clip (30–60s)
2. **Step 1:** Transcribe it with Whisper
3. **Step 2:** Summarize or extract key sentences
4. **Step 3:** Use Stable Diffusion to generate 1–2 visuals
5. **Output:** Show final images alongside the original audio snippet

Tools You'll Need

Task	Tool
Audio to Text	Whisper (OpenAI), HuggingFace ASR
Text to Visual	Stable Diffusion (via HuggingFace or Replicate)
UI Mockup	Streamlit, Flask, or simple Jupyter notebook

Prototype git link: <https://github.com/mohammadmujahiddin/kuku>